

Electronics notes:-

1. In solving diode circuits with multiple diodes, *first always assume that both diodes are conducting, verify it using KVL, KCL*. Ideally, *when a diode is on, current flows from p to n* (as diode is a unidirectional device). If supposition is wrong assume another way....

2. *Zener diode will work only if $V > V_{br}$ in Reverse bias* (breakdown voltage)

More about zener:-

(i) When *minimum power rating is asked or maximum power rating is asked* , both mean *the same thing* , It asks the $P_z (\text{max}) = I_z (\text{max}) \times V_z$, when breakdown occurs

Assume $I_r (\text{min})$ zero when not given anything.

(ii) When R_L is variable (Load Variation)

$I_{in} (\text{fixed}) = I_z (\text{variable}) + I_r (\text{variable})$ (By KCL)

So, I_r is minimum when I_z is Maximum and I_r is maximum when I_z is minimum

(iii) When V_{in} is variable (Line Variation)

$I_{in} (\text{variable}) = I_z (\text{variable}) + I_r (\text{fixed})$ (By KCL)

I_{in} is minimum when, I_z is minimum and maximum when I_z is maximum.

So, I_r is minimum when I_z is Maximum and I_r is maximum when I_z is minimum

3. To find minimum value of R_L or V_{in} in a Zener diode circuit, assume $I_z = 0$ (or simply remove the zener diode from the circuit) and $V = V_z$

To find maximum value of R_L or V_{in} in a Zener diode circuit, assume $V = V_z$ and $I_z = I_z [\text{max}]$ (and apply junction law)

4. Transfer Characteristics means graph of V_{out} vs. V_{in} .

5. $n_i = B T^{3/2} e^{(-E_g/2kT)}$

B is a material dependent parameter

B (for silicon) = $7.3 \times 10^{15} \text{ cm}^{-3} \text{ K}^{-3/2}$

E_g (for silicon) = 1.12 eV

$k = 8.62 \times 10^{-5} \text{ eV}$

Or simply remember *at room temperature $T = 300\text{ K}$ *,

$$*kT = 0.026\text{ eV}*$$

6. For silicon,

$$\text{Electron mobility} = 1400\text{ cm}^2/\text{Vs}$$

$$\text{Hole mobility} = 450\text{ cm}^2/\text{Vs}$$

7. For any semiconductor,

$$N_d + p = N_a + n \text{ (Charge neutrality)}$$

8. *Einstein's relationship*:-

$$D/(\mu) = V_t = kT/q$$

V_t = Thermal voltage

9. Since collision of 3 particles is less likely to occur, indirect band gap semiconductor (like Si) has very less probability of e-h recombination. So, Si can't be used in LEDs. Instead, GaAs is used.

10. *Derivation of contact potential*

$$*V_o = V_t \ln(N_a N_d / n_i^2)*$$

$$V_t = kT/q = 0.026\text{ V}$$

$$J_p(\text{net}) = 0$$

Use Einstein's relationship.

$$\text{And use } E = -dV/dx$$

integrate both sides and

Obtain relation in terms of N_a , N_d , n_i

Value of V_o is 0.7 V for Si and 0.3 V for Ge

11. *Derivation of Width of depletion layer (space charge)*

Assumptions:-

- a. Doping is uniform
- b. All atoms are ionised in depletion layer
- c. Electric field is confined to the junction region only and vanishes in the quasi neutral regions.

Use Poisson's equation - (from Gauss law)

$$\frac{d^2V}{dx^2} = -\rho/\epsilon$$

(Charge in depletion region on P side is (-ve))

Obtain Electric field by integrating and putting boundary conditions , separately for region R1 and R2,

Use conservation of charge to obtain a relationship between x_n and x_p

$$\text{Width } W = x_n + x_p$$

Obtain x_n in terms of W (i)

Obtain V for R1 assuming $-x_n$ as reference and for R2 using the fact that at $x=0$, both expressions will give the same value.

$$\text{Now, contact potential} = \text{Area of } E\text{-}x \text{ curve} = (1/2) \times E_{\text{max}} \times W$$

Use (i) to obtain final expression.

$$*W = \sqrt{\frac{2\epsilon V_0}{q} \left\{ \frac{1}{N_A} + \frac{1}{N_D} \right\}} *$$

12. Width of depletion region is determined by the lowest doped region .

13. In derivation of I-V characteristics:-

First write relation between $n(n_0)$ and $n(p_0)$...(i)

Only minority carriers are affected by injection.

Then write relation between $n(n_0)$ and $n(p)$(ii)

From (i) & (ii)

We obtain

$$n(p) = n(p_0) e^{(qV/kT)}$$

$$\Delta n = n(p_0) \{ e^{(qV/kT)} - 1 \}$$

Then, use *equation of continuity*

The excess charge density of holes in n region decays exponentially with distance

Then use formula for diffusion current density J_p

Find $J_{total} = J_p + J_n$

$$*J = J_s (e^{(qV/\eta kT)} - 1) *$$

Or

$$*J = J_s (e^{(V/\eta V_t)} - 1) *$$

$\eta = 1$ for Ge and 2 for Si (at low voltage)

14. Junction Capacitance (or Depletion Capacitance) is seen only dominant in Reverse Bias

$$15. *Ripple\ factor* = \frac{\sqrt{I_{rms}^2 - I_{dc}^2}}{I_{dc}}$$

16. For a full wave rectifier, We write

$$I_{dc} = C \frac{dV}{dt}$$

Or

$$I_{dc} = C V_r / T$$

Or

$$I_{dc} = 2 C V_r f$$

Where V_r = Peak to peak voltage

17. R_{ac} or ac resistance is used only when $dV/dI \rightarrow 0$ (*SMALL SIGNAL RESISTANCE*)

$$R_{ac} = V_t / I_d$$

18. Higher the value of stability factor (or closer it is to 1), the less is the stability.

19. In transistors, we assume a constant V_{be} model , means V_{be} has a drop of 0.7 V but we can't say anything about drop of V_{cb}

20. When Beta is large, $I_e = I_c$.

However, point #19 still follows

21. *When 2 terminals of a BJT are shorted, it acts as a diode*

22. In OpAmp circuits, only negative feedback is considered for closed loop gain (as it is more dominating) and when more than one voltage source is present, apply superposition theorem.

23. *For an idea opamp (with infinite gain) only*, virtual ground is created otherwise, use basic definition of gain to solve the problem.

24. However, current in input terminals of OpAmp (whether ideal or non-ideal) , is always zero.

25. *Virtual ground concept is Only valid for negative feedback* because negative feedback decreases the difference voltage. (There is some chance for gain to be infinite)

So , in schmitt trigger and astable multi- vibrator, where positive feedback is used, Virtual ground concept is not applicable

25. *Virtual ground is not applicable for open loop system*

26. Virtual ground concept is not applicable for problems where both input voltages at the terminals are known

27. *Positive feedback is important only when there is external power supply provide to the opamp and opamp is non-ideal*.

And that is called a *Schmitt Trigger Circuit* and its output is always + or - V_{sat} depending on whether V_+ or V_- is greater.

28. Input offset voltage, V_{io} is the differential input voltage V_d that exists between 2 input terminals of an opamp

Without any external inputs applied. Or simply it is the amount of input voltage that should be applied between 2 input terminals to force the output voltage to zero.

29. *Binary to grey code conversion*

Write first digit from left as it is

Just keep adding 2 consecutive digits and write the answer one by one to get gray code

30. Sometimes, $2\pi f$ is written as 's'

And sometimes 'd/dt' is written as 's'

31. In BJT, $\beta = I_c/I_b$, is valid only in active region*. If V_{ce} (sat) is given, always check whether transistor works in active region or saturation region, then work accordingly.

32. In BJT, if 2 terminals of the same junction are connected by a wire, then, it serves as a PN Junction Diode.

33. See small signal analysis of BJT

34. In Boolean Algebra, if A function $F(A,B,C,D)$ is given, *A refers to MSB* (Most significant bit) while *D refers to LSB* (Least significant bit)